

Similarly, the first focal length f_1 :

$$f_1 = \overline{H_1 F_o} = -\frac{n_{i1}}{a_{12}} \quad \overline{H_1 V_1} = -\frac{(a_{11} - 1)n_{i1}}{a_{12}} = -\frac{f(n_l - 1)d_l}{n_l R_2}$$

When a lens is in air so $n_{i1} = n_{t2} = 1$

$$\begin{aligned} \frac{1}{f} &= \frac{1}{f_1} = \frac{1}{f_2} = -a_{12} = -\mathcal{D}_1 - \mathcal{D}_2 + \frac{\mathcal{D}_1 \mathcal{D}_2 d_l}{n_l} \\ &= (n_l - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} + \frac{(n_l - 1)d_l}{n_l R_1 R_2} \right] \end{aligned}$$

For a thin lens ($d_l \ll R_1$ or $|R_2|$)

$$\frac{1}{f} = (n_l - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \overline{H_1 V_1} = \overline{H_2 V_2} = 0$$

For a system with many lenses, an easier way to obtain its optical parameters is to numerically multiply together the relevant 2x2 refraction and transfer matrixes.

Let's analyze the Tessar lens in air.

The total system matrix:

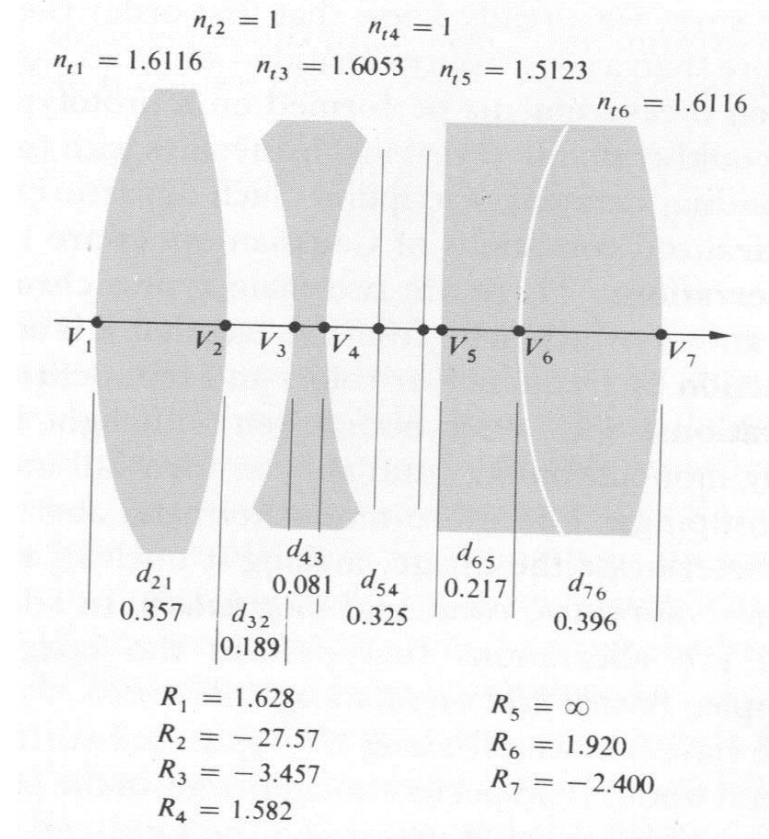
$$\mathcal{A} = \mathcal{R}_7 \mathcal{T}_{76} \mathcal{R}_6 \mathcal{T}_{65} \mathcal{R}_5 \mathcal{T}_{54} \mathcal{R}_4 \mathcal{T}_{43} \mathcal{R}_3 \mathcal{T}_{32} \mathcal{R}_2 \mathcal{T}_{21} \mathcal{R}_1$$

where

$$\mathcal{R}_i = \begin{bmatrix} 1 & -\frac{n_{right} - n_{left}}{R_i} \\ 0 & 1 \end{bmatrix}$$

$$\mathcal{T}_{i+1,i} = \begin{bmatrix} 1 & 0 \\ d_{i+1,i}/n_i & 1 \end{bmatrix}$$

n_{right}, n_{left} are the refractive indexes on the right or left side of the surface i .

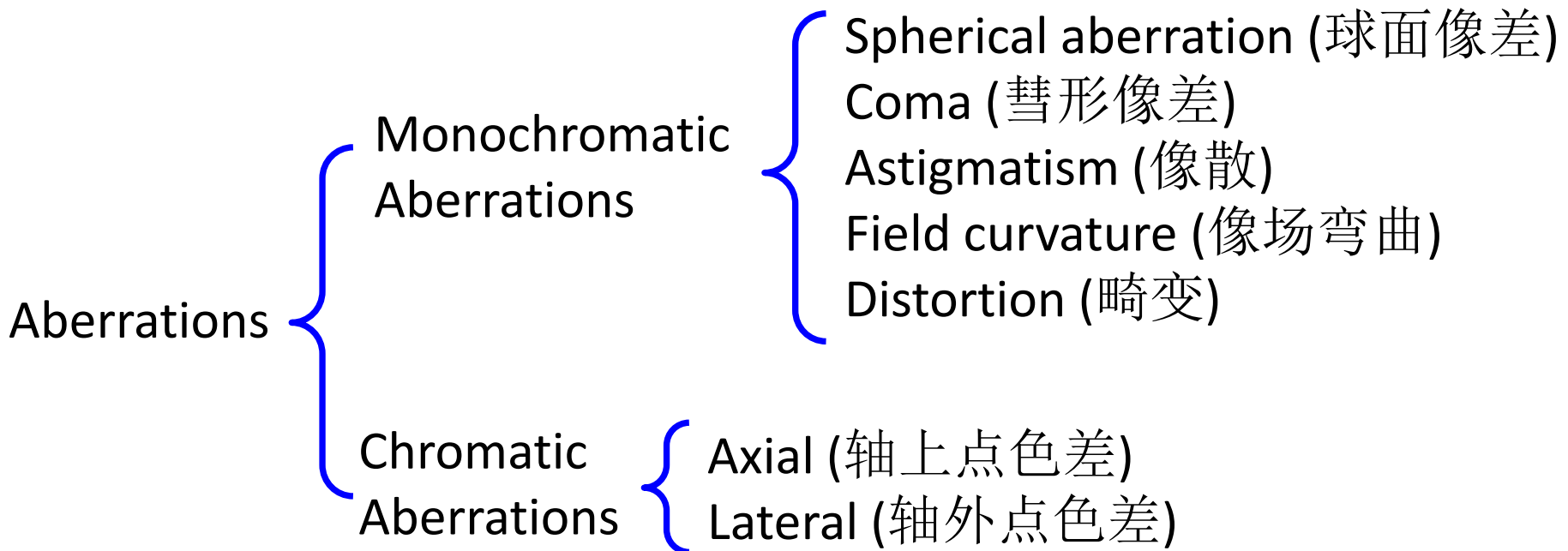


Homework

□ 6.3, 6.6, 6.18, 6.19

5.3 Aberrations

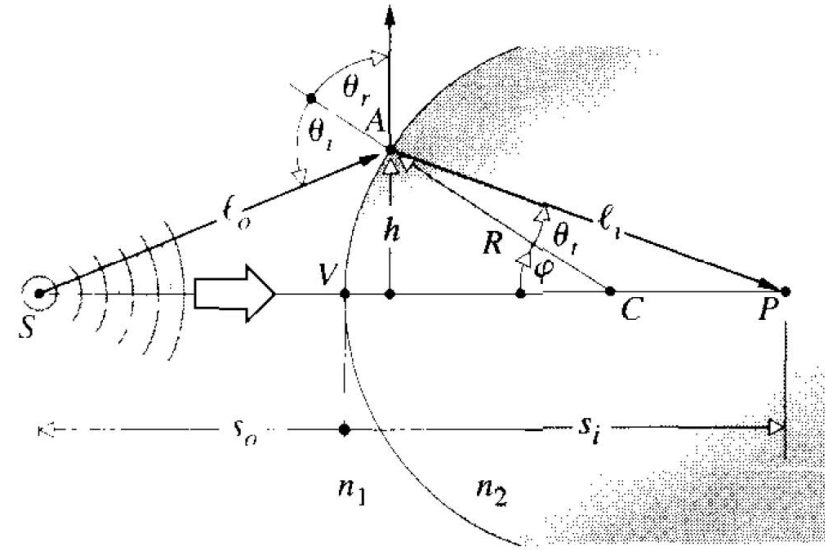
- An optical aberration is a departure of the performance of an optical system from the predictions of [Paraxial Optics \(Gaussian Optics\)](#).
- In an imaging system, it occurs when light from one point of an object does not converge into (or does not diverge from) a single point after transmission through the system.
- Aberrations occur because the simple paraxial theory is not a completely accurate model of the effect of an optical system on light (due to the wave nature of light), rather than due to flaws in the optical elements.



Monochromatic Aberrations

First-order theory:

$$\begin{aligned} \cos \varphi &= 1 - \frac{\varphi^2}{2!} + \frac{\varphi^4}{4!} - \frac{\varphi^6}{6!} + \dots \\ \sin \varphi &= \varphi - \frac{\varphi^3}{3!} + \frac{\varphi^5}{5!} - \frac{\varphi^7}{7!} + \dots \end{aligned}$$



$$\frac{n_1}{l_o} + \frac{n_2}{l_i} = \frac{1}{R} \left(\frac{n_2 s_i}{l_i} - \frac{n_1 s_o}{l_o} \right)$$

$$\begin{aligned} l_o &\approx s_o \\ l_i &\approx s_i \end{aligned}$$

$$\frac{n_1}{s_o} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R}$$

The system was restricted to operating in an extremely narrow region.

If rays from the periphery of a lens are to be included in the formation of an image.

$\sin \varphi = \varphi$: unsatisfactory

$n_i \theta_i = n_t \theta_t$: unsatisfactory

Third-order theory:

$$\begin{aligned}\cos \varphi &= 1 - \frac{\varphi^2}{2!} + \frac{\varphi^4}{4!} - \frac{\varphi^6}{6!} + \dots \\ \sin \varphi &= \varphi - \frac{\varphi^3}{3!} + \frac{\varphi^5}{5!} - \frac{\varphi^7}{7!} + \dots\end{aligned}$$

First-order theory:

$$\begin{aligned}\cos \varphi &= 1 - \frac{\varphi^2}{2!} + \frac{\varphi^4}{4!} - \frac{\varphi^6}{6!} + \dots \\ \sin \varphi &= \varphi - \frac{\varphi^3}{3!} + \frac{\varphi^5}{5!} - \frac{\varphi^7}{7!} + \dots\end{aligned}$$

Deviation between First- and Third-order theory.

Monochromatic aberrations

- ☐ Spherical aberration
- ☐ Coma
- ☐ Astigmatism
- ☐ Field curvature
- ☐ Distortion

**Primary aberrations
(Seidel aberrations)**

Spherical aberration (SA)

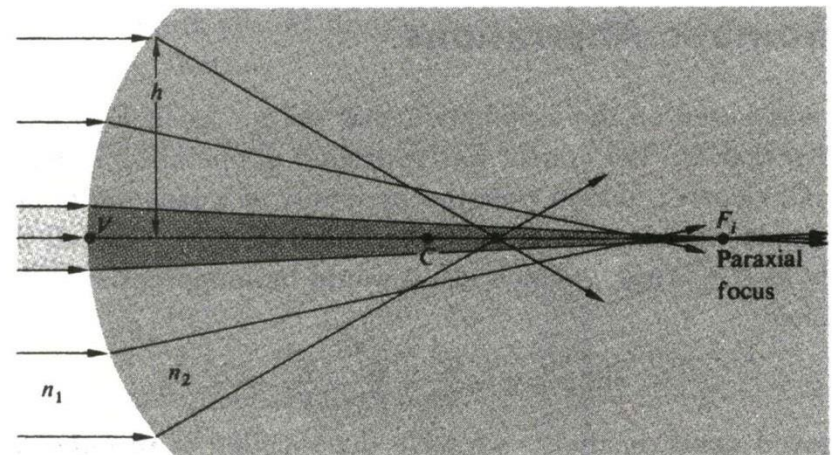
In the paraxial region
$$\frac{n_1}{s_o} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R}$$

The third-order theory: $\cos \varphi = 1 - \varphi^2 / 2$

$$\frac{n_1}{s_o} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R} + h^2 \left[\frac{n_1}{2s_o} \left(\frac{1}{s_o} + \frac{1}{R} \right)^2 + \frac{n_2}{2s_i} \left(\frac{1}{R} - \frac{1}{s_i} \right)^2 \right]$$

Meaning: rays striking the surface at greater distances above the axis (h) are focused nearer the vertex.

For non-paraxial rays, f is also related to size of optical lens.



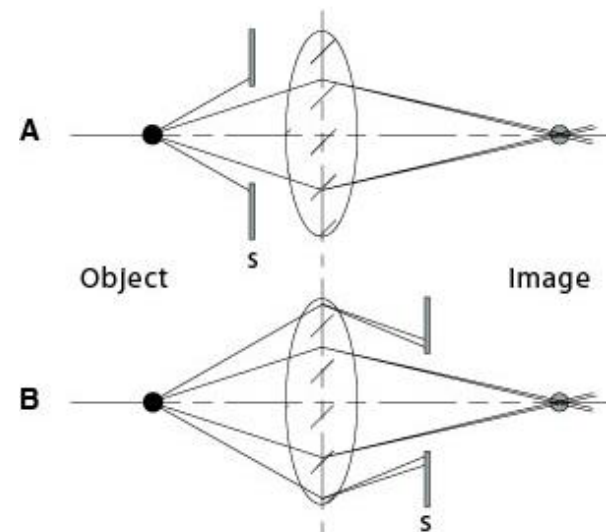
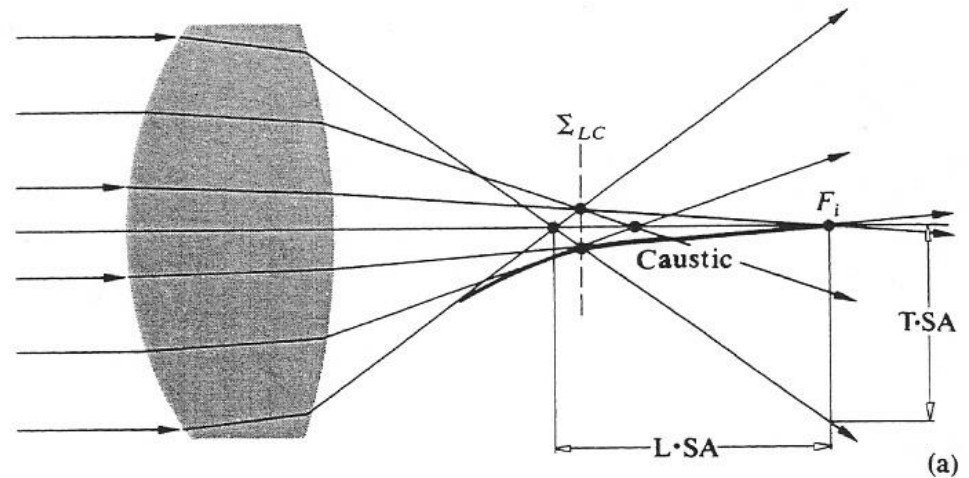
Similarly, for a converging lens, the marginal rays will be bent too much, being focused in front of the paraxial focus.

Longitudinal spherical aberration (L.SA): distance between the axial intersection of a ray and the paraxial focus.

Traverse spherical aberration (T.SA): the height above the axis where a given ray strikes screen placed at focus.

Circle of least confusion: the position where image blur will have its smallest diameter.

Correction: SA can be reduced by stopping down the aperture; but that reduces the amount of light entering the system as well.



Correction: for an object at ∞ , the round side of lens facing the object will suffer a minimum amount of SA.

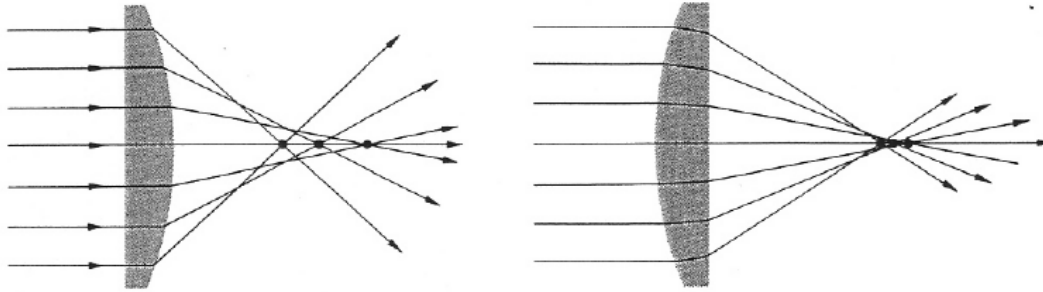


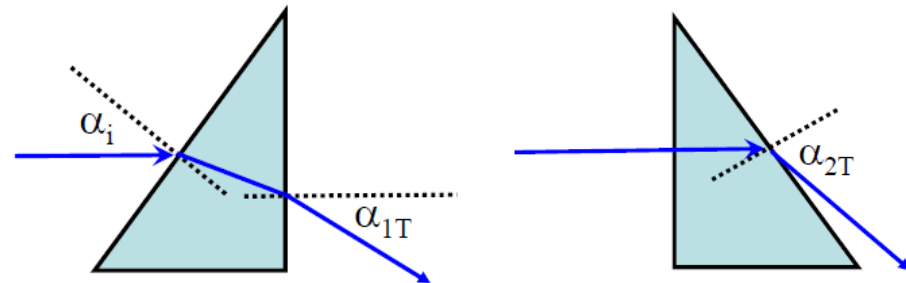
Fig. 6.16 Spherical Aberration for a planar-convex lens in both orientations.

A planar-convex lens can be approximated as two prisms

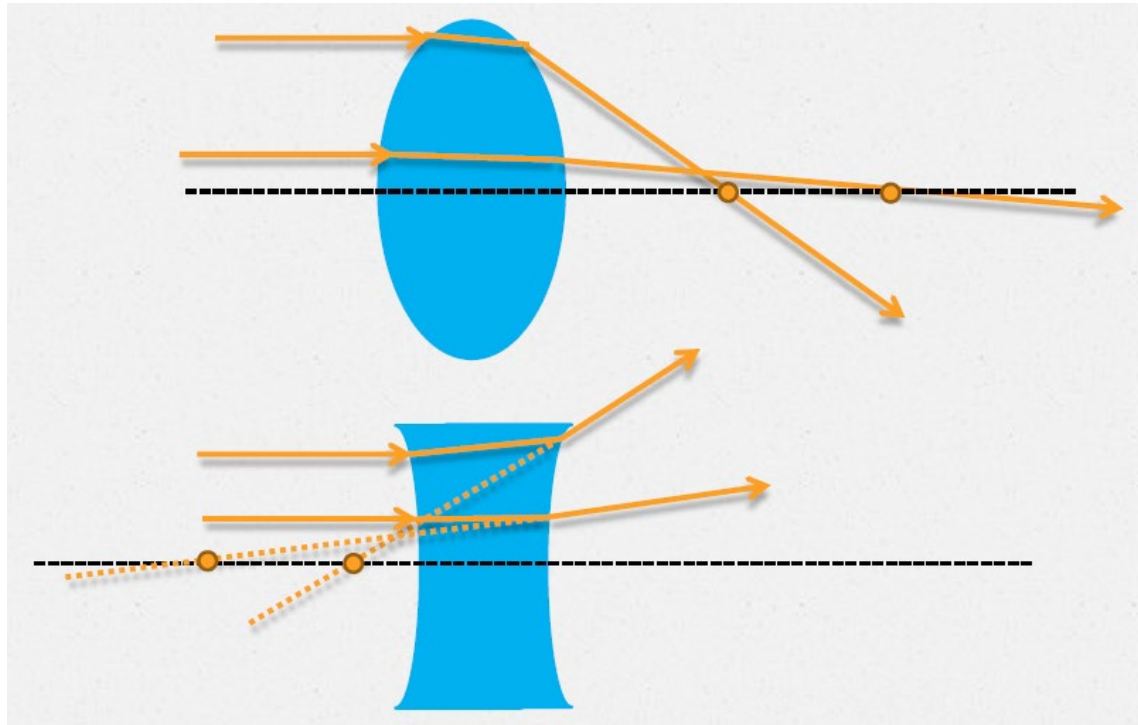


Dispersing prism: incident ray will undergo a minimum deviation when $\alpha_i \approx \alpha_t$.

$\alpha_{2t} > \alpha_{1t}$: the right prism results in a greater deviation.

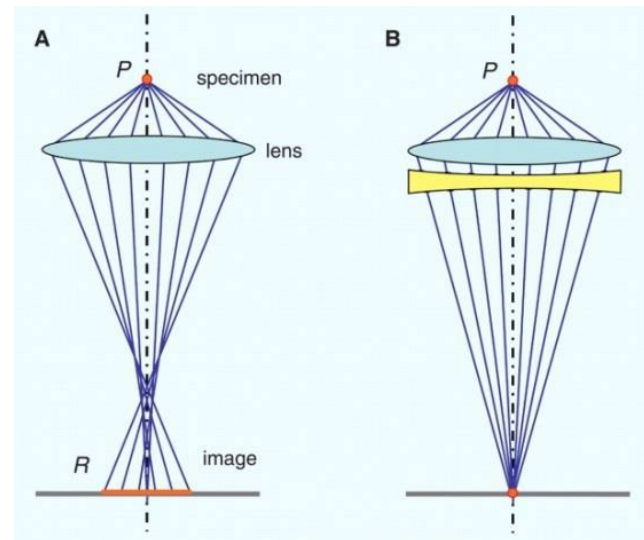


For positive lens
 $L.SA > 0$



For negative lens
 $L.SA < 0$

Correction: combination of
positive and negative lens
elements with different thickness

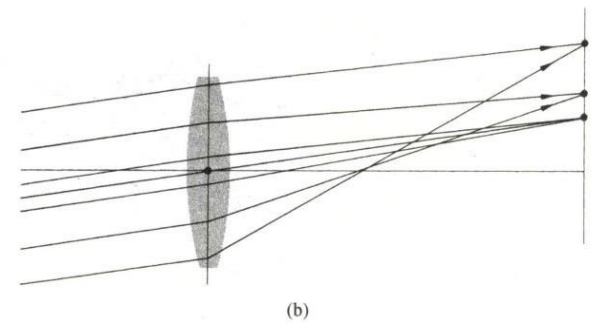
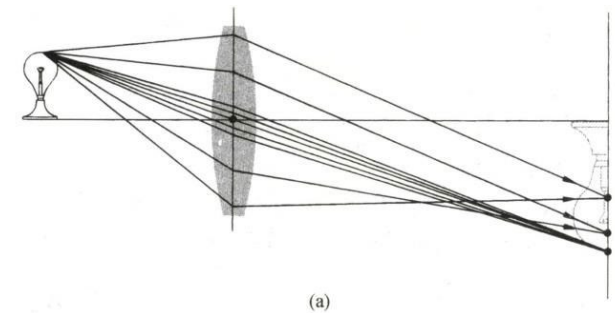
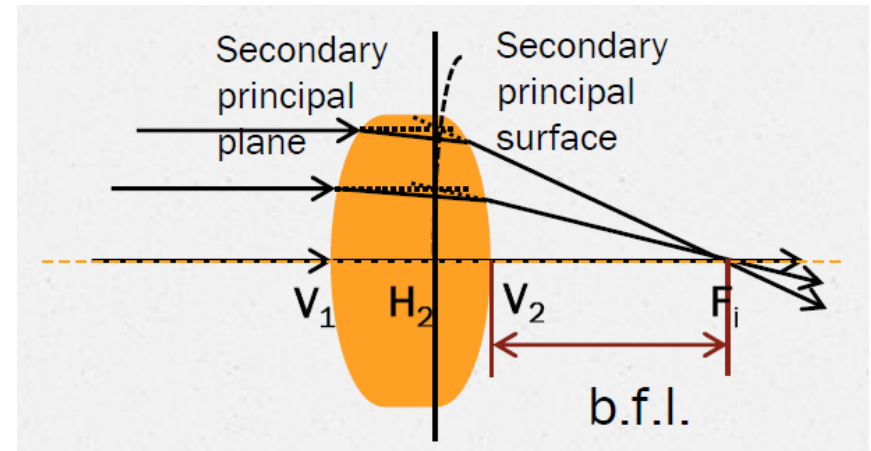


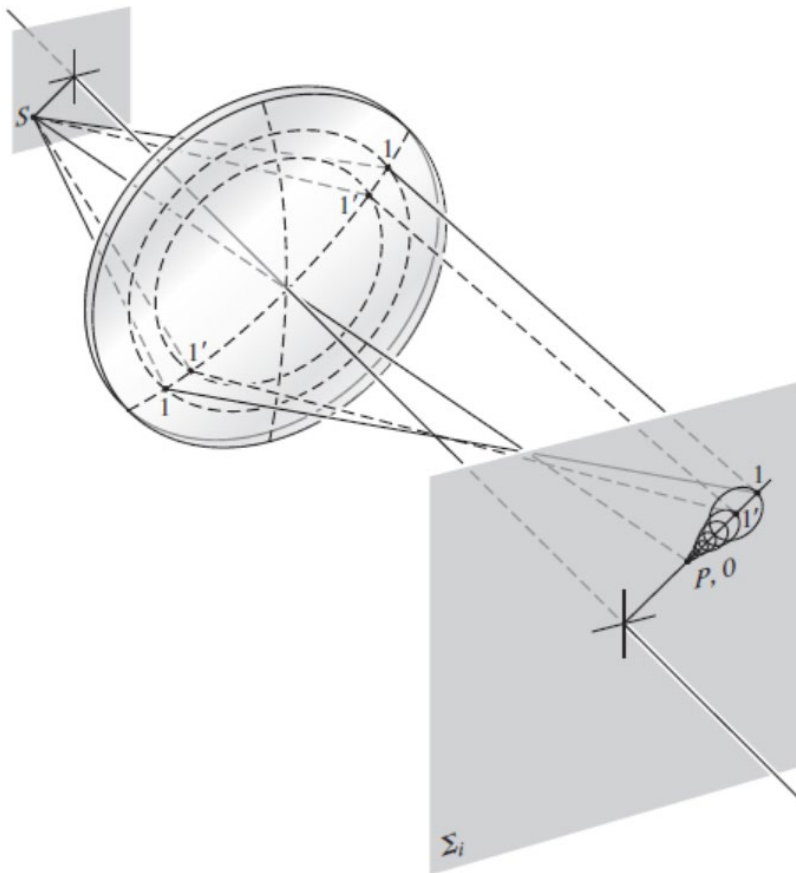
Coma (comatic aberration)

Coma is associated with the fact that the principal planes are really curved surfaces resulting in a different M_T for both marginal and central rays.

Since $M_T = -\frac{f}{x_o} = -\frac{x_i}{f}$, the curved nature of the principal surface will result that the effective focal lengths, therefore the transverse magnifications will differ for rays traversing off-axis regions of the lens.

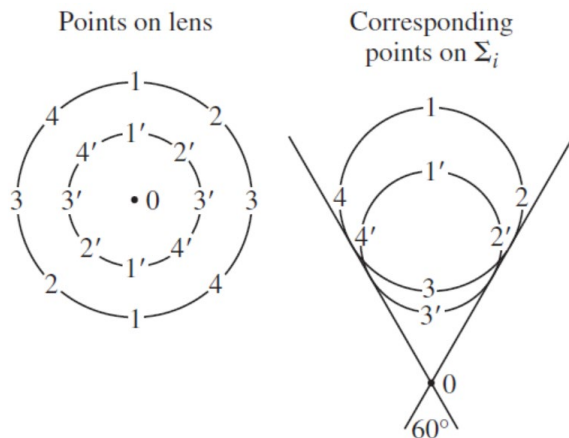
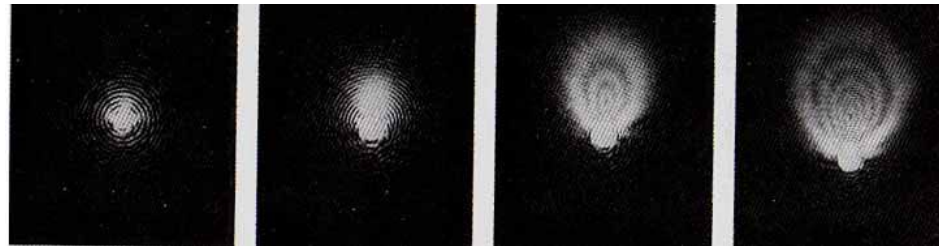
The variation in M_T depends on the location of the object which can result in a negative (a) or positive coma (b).





The imaging of a point at S can result in a “comet-like” tail, known as a coma flare and forms a “comatic” circle on the screen Σ (positive coma in this case). This is often considered the worst out of all the aberrations, primarily because of its asymmetric configuration.

Coma effect



For the outmost rays

Tangential coma (切向慧差)

Sagittal coma (弧矢慧差)

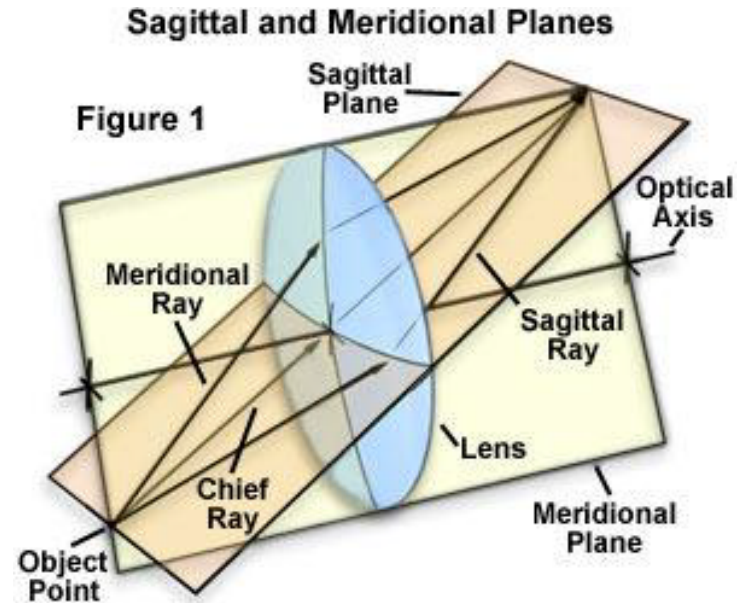
Astigmatism

Meridional plane (tangential plane) (子午面): contains both the chief ray and the optical axis

Sagittal plane (弧矢面): contains the chief ray and is perpendicular to the meridional plane.

Fermat's principle

Planes containing the tilted rays give a shorter focal length which depends on the power of the lens and the angle of inclination. The result is that there are both a meridional focus F_t and a sagittal focus F_s .



Tilted rays have a shorter focal length.

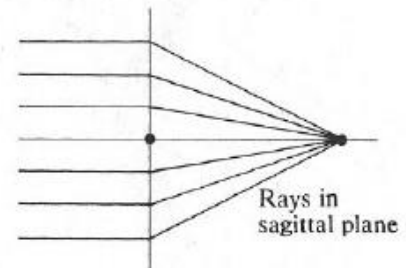
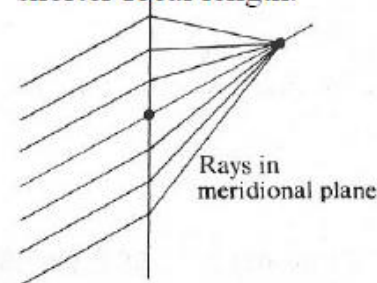
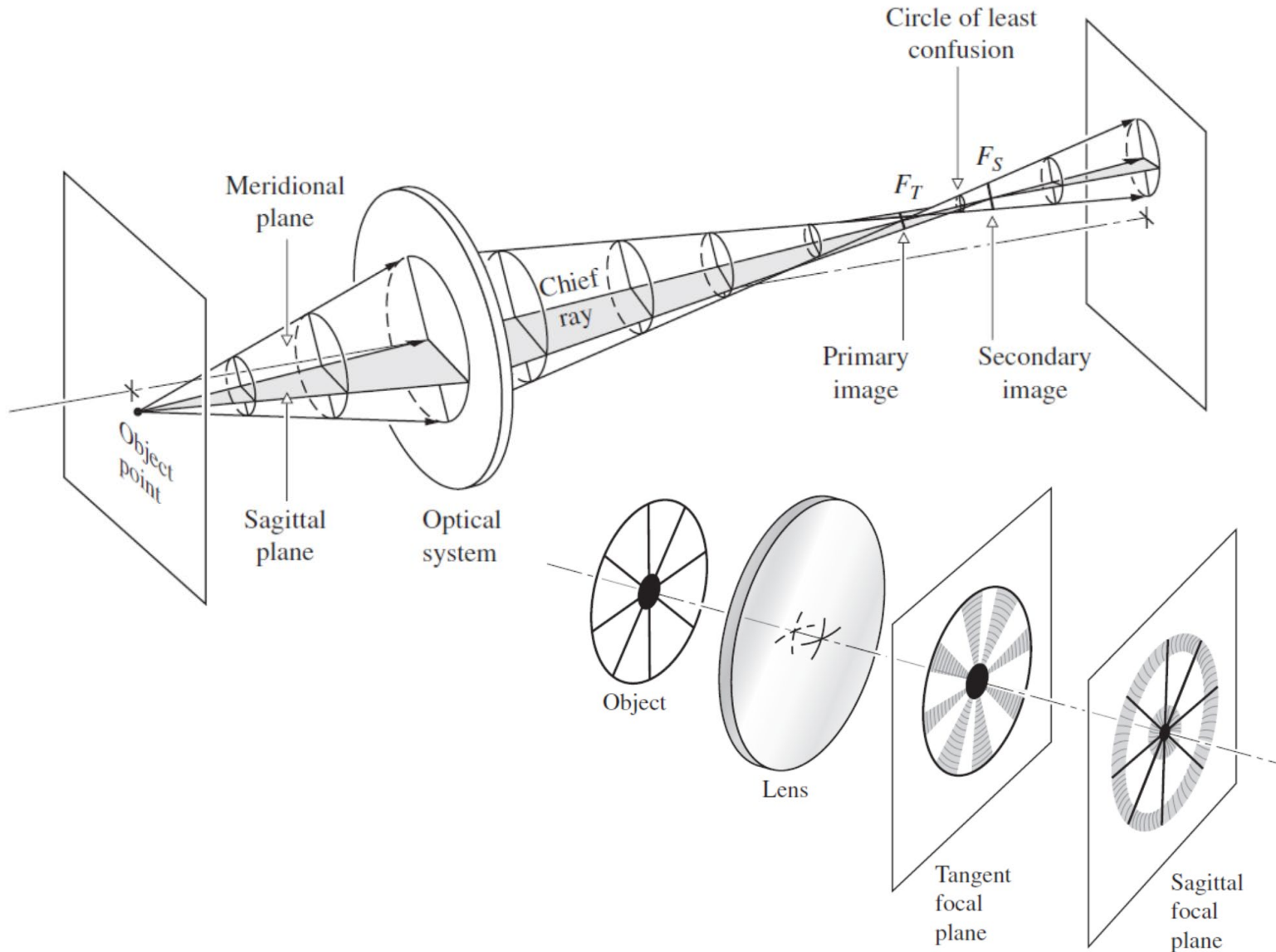


Figure 6.26 The sagittal and meridional planes.

Astigmatism: Note that the cross-section of the beam changes from a circle $\rightarrow$ ellipse $\rightarrow$ line (primary image) $\rightarrow$ ellipse $\rightarrow$ circle of least confusion $\rightarrow$ ellipse $\rightarrow$ line (secondary image) .



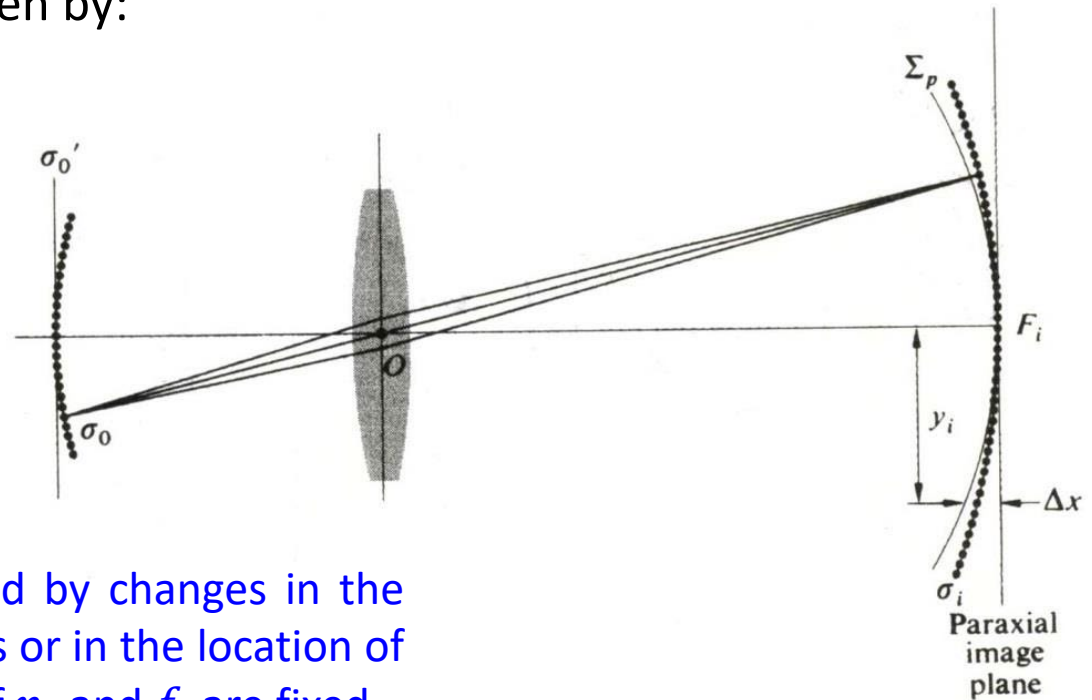
Field curvature

Petzval field curvature: a planar object normal to the axis will be imaged curve stigmatic image surface.

- ❑ A positive lens : it curves inward toward the object plane
- ❑ A negative lens: it curves outward away from that plane.

The displacement of an image point at height on the Petzval surface from the paraxial image plane is given by:

$$\Delta x = \frac{y_i^2}{2} \sum_{j=1}^m \frac{1}{n_j f_j}$$



Petzval surface will be unaltered by changes in the positions or shapes of the lenses or in the location of the stop, as long as the values of n_j and f_j are fixed.

Petzval condition

In case of two thin lenses having any space:

Δx can be made zero provided that:

$$\frac{1}{n_1 f_1} + \frac{1}{n_2 f_2} = 0$$

If $f_1 = -f_2$ and $n_1 = n_2$ (combine a positive and a negative thin lens together)

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

$$f = \frac{f_1^2}{d}$$

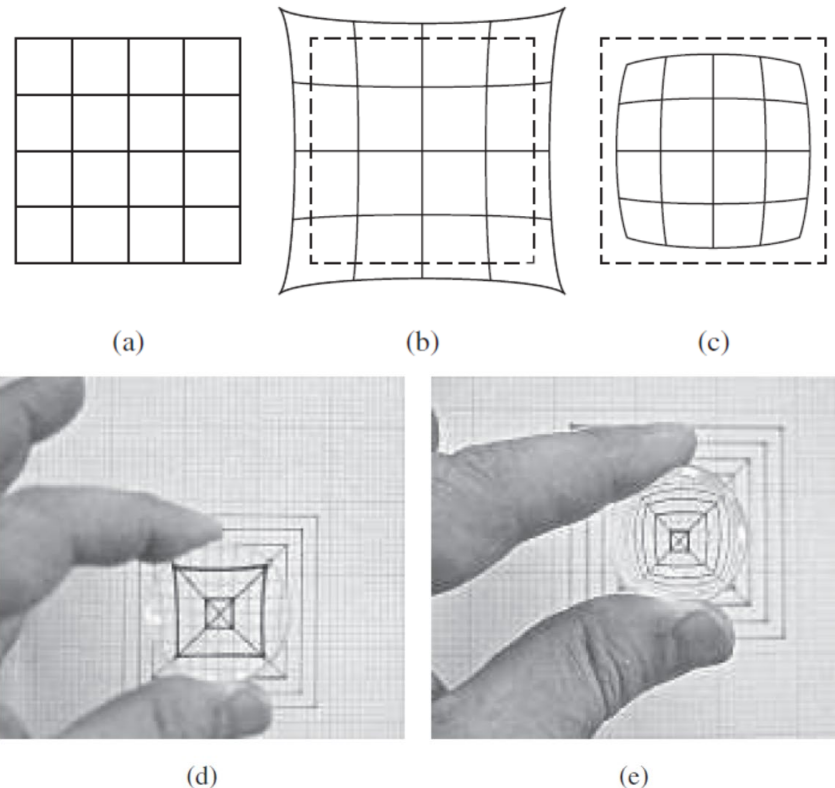
A flat field and a finite positive focal length.

Distortion

Origin: the transverse magnification (M_T) is a function of the off-axis image distance (y_i). In other words, different areas of the lens have different focal lengths and different magnifications.

Result: misshaping of the image

- ❑ Positive or pincushion distortion: each image point is displaced radially outward from the center with the most distant points moving the greatest amount (M_T increases with y_i)
- ❑ Negative or barrel distortion: each point on the image moves radially inward toward the center (M_T decreases with y_i)

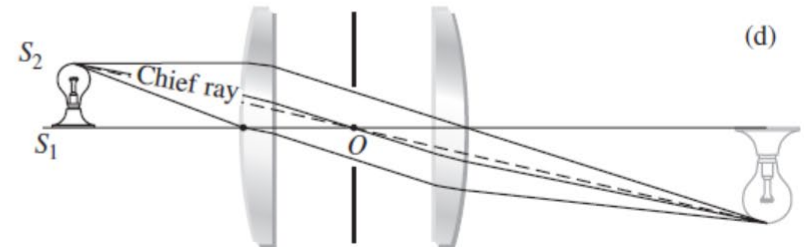
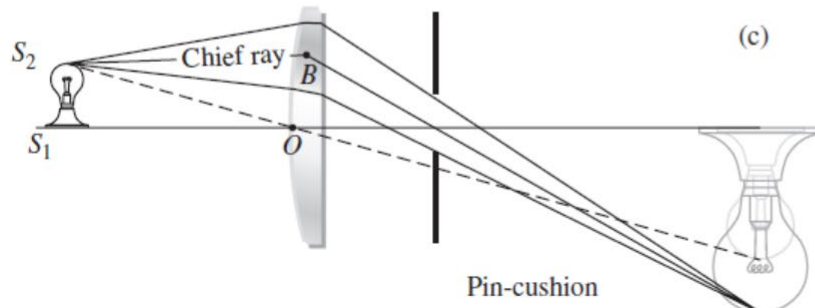
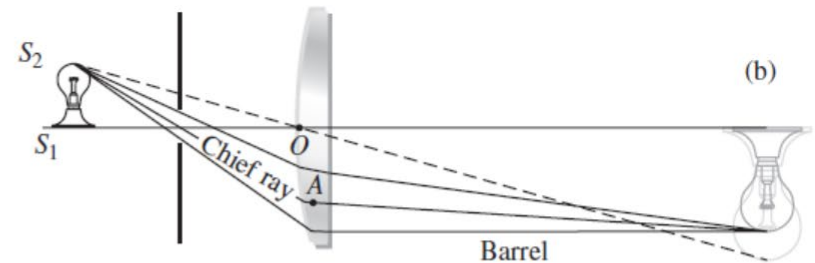
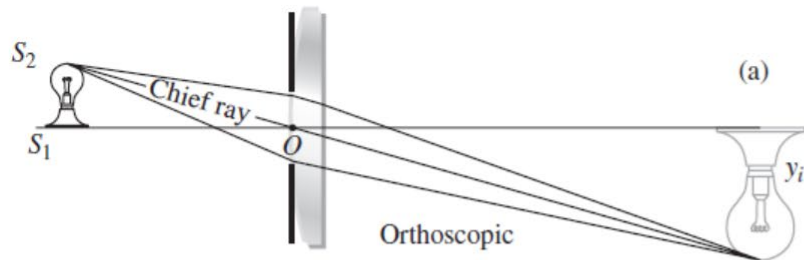


The effect of stop location on distortion

The stop in front of a positive lens: $S_2A > S_2O$ thus x_o will be greater and M_T will be smaller. **Barrel distortion**

The stop behind a positive lens: $S_2B < S_2O$ thus x_o will be decreased and M_T will be increased. **Pincushion distortion**

$$M_T \equiv \frac{|y_i|}{y_o} = -\frac{x_i}{f} = -\frac{f}{x_o}$$



Chromatic Aberrations

Thin-lens equation

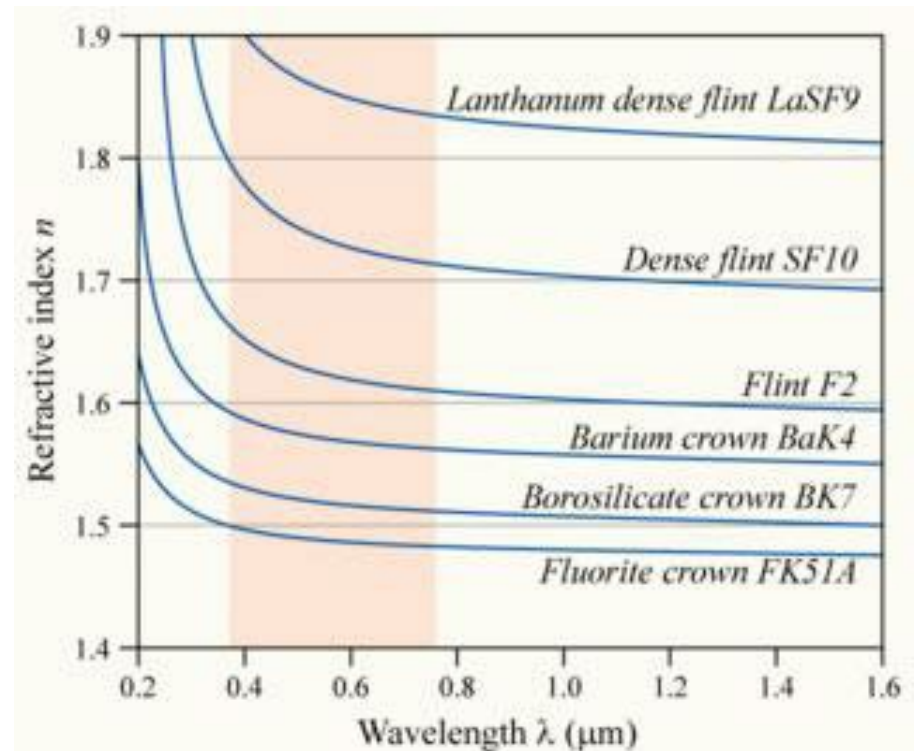
$$\frac{1}{f} = (n_l - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

where n_l is wavelength-dependent

Note: the focal length depends on the wavelength

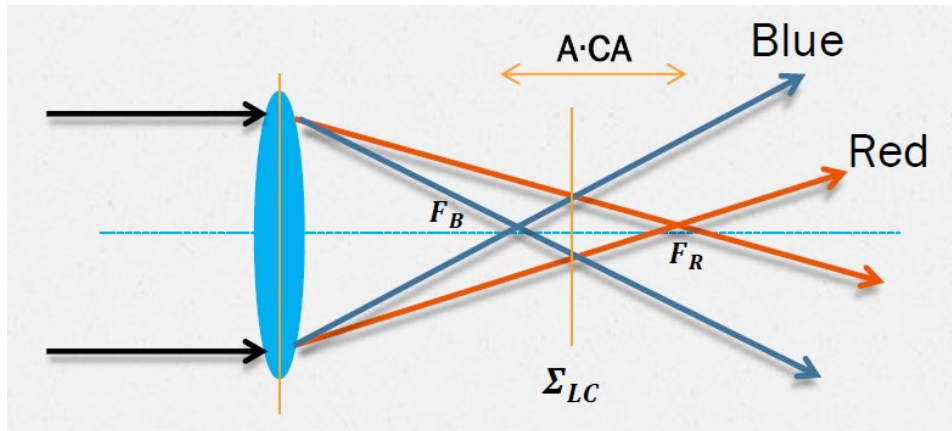
In general, $n_l(\lambda)$ decreases with wavelength over the visible region, and thus $f(\lambda)$ increases with λ .

Example: $f_{red} > f_{blue}$



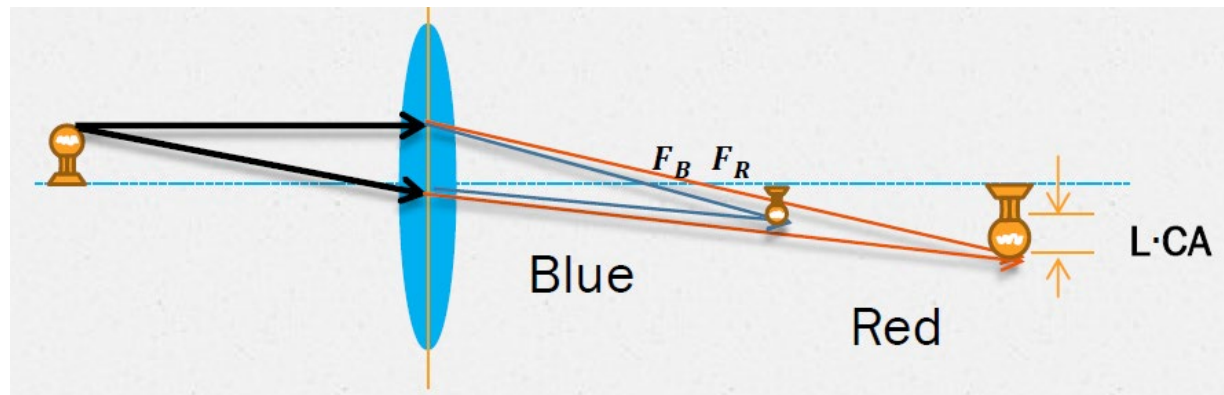
A positive lens

Axial chromatic aberration (A.CA): the axial distance between two focal points spanning a given frequency range.



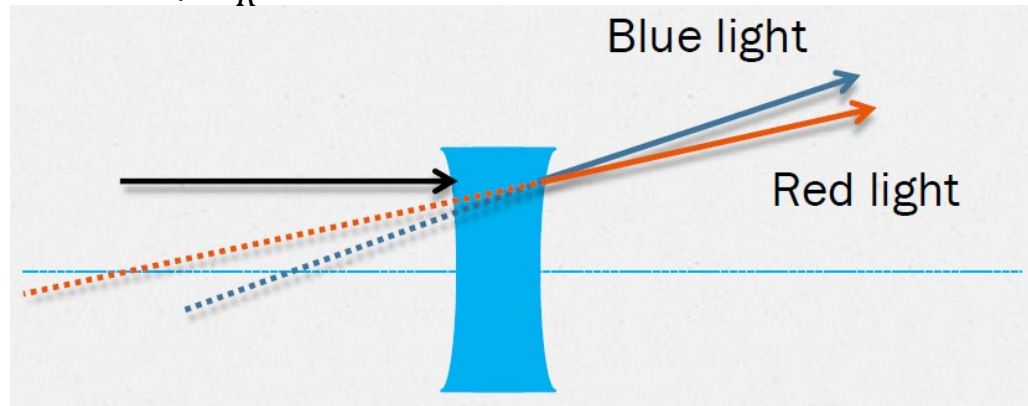
The blue focus, F_B , is to the left of the red focus, F_R , the A.CA is positive.

Lateral chromatic aberration (L.CA): the image of an off-axis point will be formed of the constituent frequency components, each arriving at a different height above the axis.



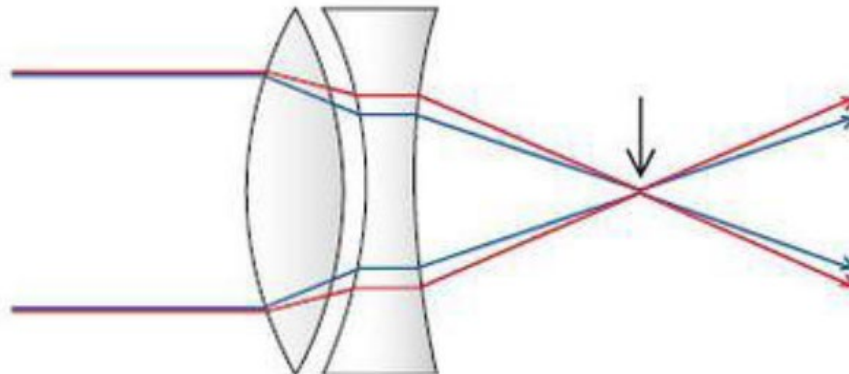
A negative lens

A negative lens generates negative A.CA, which means The blue focus, F_B , is to the right of the red focus, F_R .



Thin achromatic doubles

A combination of two thin lenses, one positive and one negative, could conceivably result in the precise overlapping of F_R and F_B .



Thin achromatic doubles

With two lenses separated by a distance d , the effective focal length:

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

When the focal length of the doublet for red (f_R) and blue (f_B) light are equal for contact lens ($d=0$), the focal length of the compound lens for yellow (f_Y) light, roughly midway between the blue and red extremes, satisfies the following equation.

$$\frac{f_{2Y}}{f_{1Y}} = - \frac{(n_{2B} - n_{2R}) / (n_{2Y} - 1)}{(n_{1B} - n_{1R}) / (n_{1Y} - 1)}$$

Dispersive powers: $\frac{n_{2B} - n_{2R}}{n_{2Y} - 1}$ and $\frac{n_{1B} - n_{1R}}{n_{1Y} - 1}$

Dispersive indices (Abbe numbers) V : $\frac{1}{\text{dispersive powers}}$

We have $f_{1Y}V_1 + f_{2Y}V_2 = 0$

For practical application, the specific spectral lines whose wavelengths are known with great precision is applied

TABLE 6.1 Several Strong Fraunhofer Lines

Designation	Wavelength (Å)*	Source
<i>C</i>	6562.816 Red	H
<i>D</i> ₁	5895.923 Yellow	Na
<i>D</i>	Center of doublet 5892.9	Na
<i>D</i> ₂	5889.953 Yellow	Na
<i>D</i> ₃ or <i>d</i>	5875.618 Yellow	He
<i>b</i> ₁	5183.618 Green	Mg
<i>b</i> ₂	5172.699 Green	Mg
<i>c</i>	4957.609 Green	Fe
<i>F</i>	4861.327 Blue	H
<i>f</i>	4340.465 Violet	H
<i>g</i>	4226.728 Violet	Ca
<i>K</i>	3933.666 Violet	Ca

*1 Å = 0.1 nm.

Thin achromatic doubles

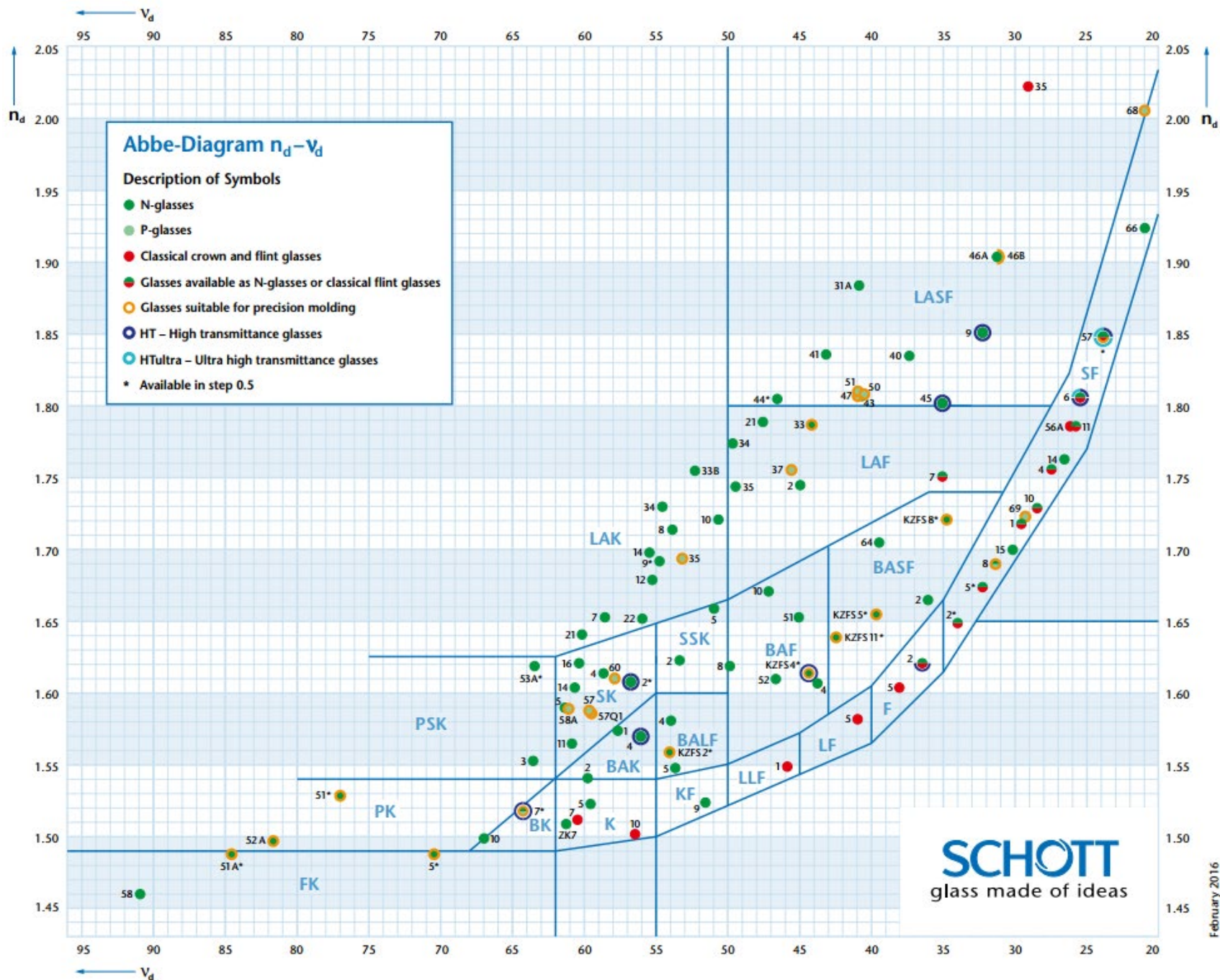
$$f_{1d}V_{1d} + f_{2d}V_{2d} = 0$$

$$V_d = \frac{n_d - 1}{n_F - n_C}$$

TABLE 6.2 Optical Glass

Type Number	Name	n_D	V_D
511:635	Borosilicate crown—BSC-1	1.5110	63.5
517:645	Borosilicate crown—BSC-2	1.5170	64.5
513:605	Crown—C	1.5125	60.5
518:596	Crown	1.5180	59.6
523:586	Crown—C-1	1.5230	58.6
529:516	Crown flint—CF-1	1.5286	51.6
541:599	Light barium crown—LBC-1	1.5411	59.9
573:574	Barium crown—LBC-2	1.5725	57.4
574:577	Barium crown	1.5744	57.7
611:588	Dense barium crown—DBC-1	1.6110	58.8
617:550	Dense barium crown—DBC-2	1.6170	55.0
611:572	Dense barium crown—DBC-3	1.6109	57.2
562:510	Light barium flint—LBF-2	1.5616	51.0
588:534	Light barium flint—LBF-1	1.5880	53.4
584:460	Barium flint—BF-1	1.5838	46.0
605:436	Barium flint—BF-2	1.6053	43.6
559:452	Extra light flint—ELF-1	1.5585	45.2
573:425	Light flint—LF-1	1.5725	42.5
580:410	Light flint—LF-2	1.5795	41.0
605:380	Dense flint—DF-1	1.6050	38.0
617:366	Dense flint—DF-2	1.6170	36.6
621:362	Dense flint—DF-3	1.6210	36.2
649:338	Extra dense flint—EDF-1	1.6490	33.8
666:324	Extra dense flint—EDF-5	1.6660	32.4
673:322	Extra dense flint—EDF-2	1.6725	32.2
689:309	Extra dense flint—EDF	1.6890	30.9
720:293	Extra dense flint—EDF-3	1.7200	29.3

From T. Calvert, "Optical Components," *Electromechanical Design*, May 1971. Type number is given by $(n_D - 1)(10 V_D)$, where n_D is rounded off to three decimal places. For more data, see Smith, *Modern Optical Engineering*, Fig. 7.5.



Example: design a Fraunhofer achromat of focal length 50 cm. It is formed of a crown double-convex in contact with a concave-planar flint lens.

Solution:

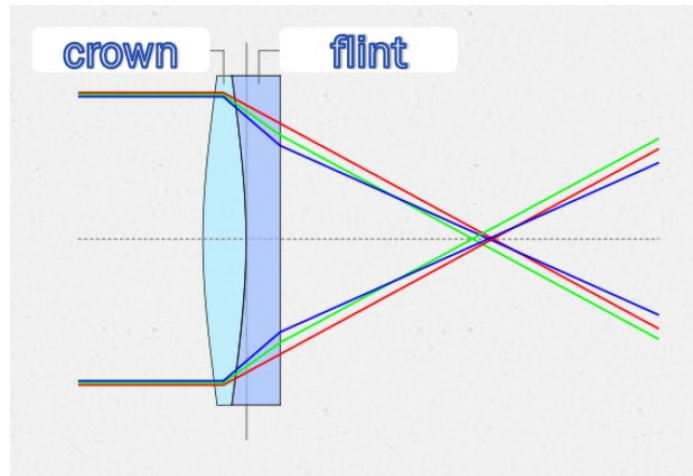
Thin achromatic doublets

$$f_{1d}V_{1d} + f_{2d}V_{2d} = 0$$

$$V_d = \frac{n_d - 1}{n_F - n_C}$$

The effective focal length:

$$\frac{1}{f_d} = \frac{1}{f_{1d}} + \frac{1}{f_{2d}}$$



For the derivation, please refer to textbook.

Separated achromatic doublets

Achromatize the focal length of a doublet composed of two widely separated elements of the same glass.

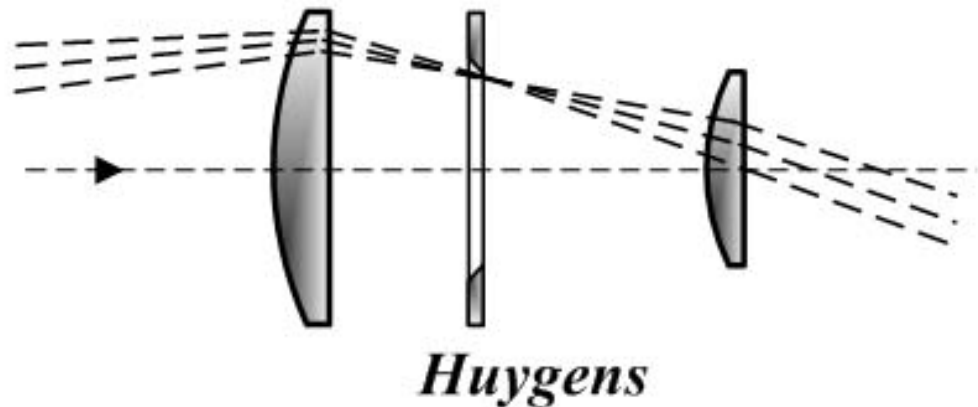
$$n_{1R} = n_{2R} = n_R \qquad n_{1B} = n_{2B} = n_B$$

The separation between two lens:

$$d = \frac{f_{1Y} + f_{2Y}}{2}$$

In d-light

$$d = \frac{f_{1d} + f_{2d}}{2}$$



Homework

□ 6.26